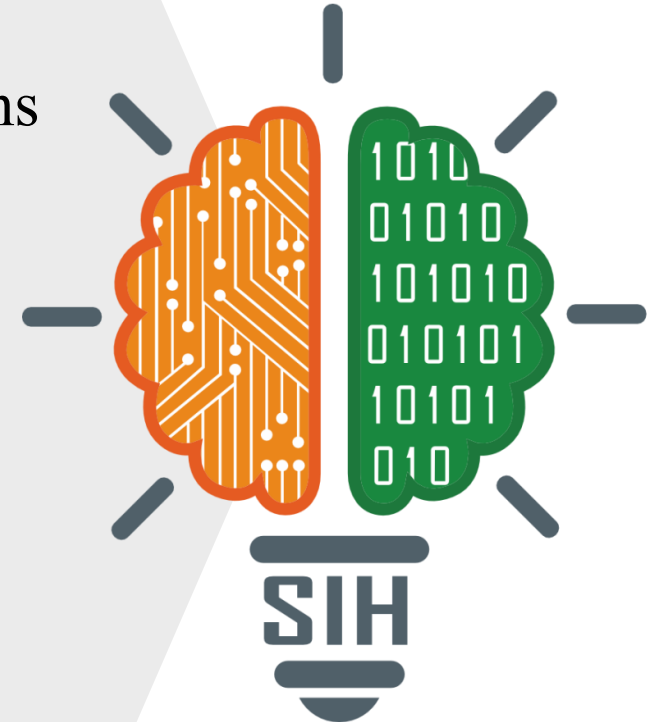


SMART INDIA HACKATHON 2026



- **Problem Statement ID - SIH26007**
- **Problem Statement Title- MineSight - Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.**
- **Theme - Smart Automation**
- **PS Category - Hardware**
- **Team Name - WeBuildZ**
- **Team ID -**



PROBLEM

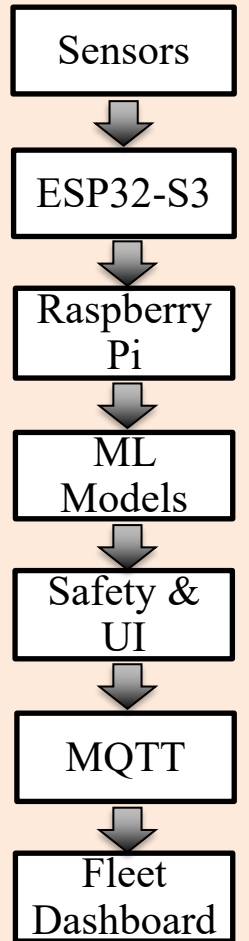
- **Safety Crisis:** 3–5m visibility vs 30m heavy vehicle braking distance leads to fatal rear-end collisions and steep bench roll-offs.
- **Economic Loss:** 50–60 days of monsoon slowdowns per year, idling multimillion-dollar fleets and stalling ore evacuation.
- **Tech Gap:** Optical cameras fail from backscatter glare; deep pit benches have zero cellular network for cloud alerts.
- **Connectivity Gap:** Cameras blind from fog glare; deep mine pits lack cellular connectivity.

Our Solutions

- **Active Safety:** Thermal-first vision (100m+ range) with in-cabin HUD road guidance and 3-stage collision auto-braking.
- **Zero Shutdowns:** Restores safe 20–30 km/h haulage, recovering 50–60 lost monsoon days and ₹8–15+ Cr annually.
- **Offline V2V Mesh:** Peer-to-peer LoRa network shares blind-spot obstacle data with zero reliance on cellular or internet.

Innovations

- **Cooperative Fog-Vision:** Peer-to-peer V2V mesh lets dumpers "see around" blind hairpin bends through other trucks' sensors.
- **Physics-Proof Sensor Fusion:** Fuses Thermal IR (penetrates fog) + LiDAR (3D geometry) + Radar (speed/rain) for 100% weather resilience.
- **Low-Cost Retrofit:** Plug-and-play modular unit costs under ₹35,000 vs. ₹16–30 Lakhs for proprietary OEM systems.



L0 Hardware



Thermal Camera



LiDAR



mmWave Radar



RTK GPS



LoRa (V2V Mesh)

L1 Controller



ESP32-S3



Raspberry Pi

L2 ML Model



OpenCV



PyTorch



TensorFlow



NumPy



Sensor Fusion



TinyML

L3 Dashboard



Next.js



WebSocket



MQTT



Three.js



Tailwind CSS



In-Cabin Dashboard



Fleet Dashboard



Technical Feasibility

Modular hardware – Thermal, LiDAR, Radar, Stereo and GPS can be integrated as independent sensor modules.

Edge processing – Raspberry Pi 5 + Coral handles AI, sensor fusion and safety decisions.

Dedicated safety controller – ESP32-S3 manages sensor interfaces, alerts, V2V communication and watchdog functions.

Open-source software – YOLO, OpenCV, Kalman Filter, MQTT and Web technologies reduce development dependency.

Fallback architecture – Basic safety can continue through ESP32 + LiDAR/Radar if the main compute unit fails.



Viability

- **Incremental deployment** – Start with **one vehicle**, validate the system, then extend to multiple dumpers.
- **Existing infrastructure** – Dashboard, V2V and sensor modules are designed as **separate layers**, making future upgrades easier.
- **Prototype cost** – Target configuration is approximately **₹25,000–₹40,000** depending on available laboratory hardware.
- **Low-cost alternatives** – Sensors can be replaced with **MLX90640, YDLIDAR, ESP-NOW and other lower-cost options**.



KEY RISKS → MITIGATION

- **V2V range limitations** → Combine **ESP-NOW + LoRa** for short- and long-range communication.
- **Environmental Risk** → Develop **AI, Web and IoT modules in parallel** and integrate incrementally.

IMPACTS

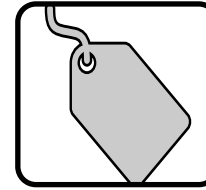
❖ on Driver (Operator)

- **Zero Fog Blindness:** Restores 100m+ clear virtual vision in 3–5m fog via the in-cabin HUD, eliminating severe driver panic and visual stress.
- **Fatal Hazard Protection:** 3-stage proactive alerts (HUD visual, seat vibration, auto-brake relay) prevent blind rear-end collisions and cliff-edge roll-offs.

❖ on Fleet Management

- **Zero Weather Shutdowns:** Recovers **50–60 lost or slowed monsoon operating days**, maintaining safe 20–30 km/h haulage during heavy fog.
- **Real-Time Fleet Oversight:** Central 3D Digital Twin gives control room operators live vehicle tracking, V2V mesh connectivity, and dynamic fog hazard heatmaps.

BENEFITS



Capital Efficiency

Retrofit BOM costs < **₹35,000 per truck** vs. ₹16–30 Lakhs for proprietary Caterpillar/Hexagon systems.



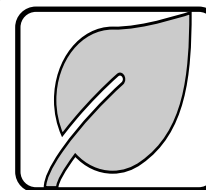
Economic Savings

Saves **₹8–15+ Crores annually** per complex by eliminating idle multimillion-dollar fleet standbys and delayed ore evacuation.



Statutory DGMS Compliance

100% compliant with Directorate General of Mines Safety (DGMS) mandatory Collision Avoidance System norms. **Statutory DGMS Compliance**



Environmental Decarbonization

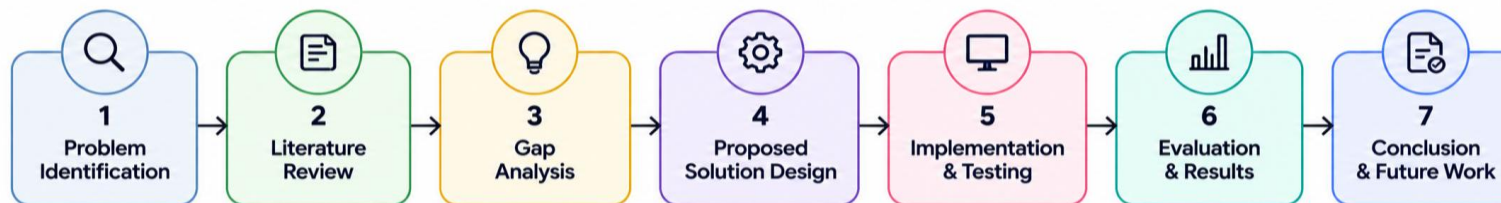
Eliminates thousands of hours of diesel engine idling during monsoon shutdowns, saving fuel and cutting CO₂ emissions.

❖ References

- 1) GitHub - [MineSight](#)
- 2) Chaulya et al. (2025) – [Research Paper](#)
- 3) Hexagon — [Vehicle Intervention System](#)

❖ Technical References

- 1) Ultralytics — [YOLOv8](#)
- 2) OpenCV — [StereoSGBM](#)
- 3) OASIS — [MQTT v5.0 Standard](#)

❖ Research Workflow -

COMPARISONS WITH EXISTING SYSTEMS			
Dimension	Commercial	Chaulya	Our Proposal
Thermal Imaging	—	✓	✓
LiDAR	✓	✗	✓
Radar	✓	✓	✓
Sensor Fusion	—	—	✓
V2V Cooperative Perception	—	✗	✓
Central Dashboard	✓	✗	✓
Road Edge	✓	✓	✓
Multi-Stage Safety	✓	—	✓
Deployable	✗	—	✓
Cost	\$25-40k	\$20-25k	\$9-10K



Live Demo – sih.webuildz.online